

**AMENDMENT NO. 2 JANUARY 2025**  
**TO**  
**IS 6419 : 1996 WELDING RODS AND BARE ELECTRODES FOR GAS SHIELDED**  
**ARC WELDING OF STRUCTURAL STEELS — SPECIFICATION**  
*( First Revision )*

*(Page 1, clause 2)* — Substitute the following for the existing:

**‘2 REFERENCES**

The standards given below contain provisions which through reference in this text, constitute provision of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards:

| <i>IS No.</i>                                   | <i>Title</i>                                                                                                                               |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| IS 228 (relevant parts)                         | Methods for chemical analysis of steels                                                                                                    |
| IS 812 (all parts)                              | Welding and allied processes — Vocabulary                                                                                                  |
| IS 1387 : 1993                                  | General requirements for the supply of metallurgical materials ( <i>second revision</i> )                                                  |
| IS 1608 (Part 1) :<br>2022/ISO 6892-1 :<br>2019 | Metallic materials — Tensile testing: Part 1 Method of test at room temperature<br>( <i>fifth revision</i> )                               |
| IS 1757 (Part 1) :<br>2020/ISO 148-1 :<br>2016  | Metallic Materials — Charpy pendulum impact test: Part 1 Test method ( <i>fourth revision</i> )                                            |
| IS 2002 : 2024                                  | Steel plate for pressure vessel for intermediate and high temperature service including boilers — Specification ( <i>fourth revision</i> ) |
| IS 2062 : 2011                                  | Hot rolled medium and high tensile structural steel — Specification ( <i>seventh revision</i> )                                            |
| IS 3039 : 2024                                  | Structural steel for ship construction — Specification ( <i>third revision</i> )                                                           |
| IS 18632 : 2024/<br>ISO 14344 : 2010            | Welding Consumables — Procurement of filler materials and fluxes’                                                                          |

*(Page 1, clause 3, line 2)* — Substitute ‘IS 812 (Part 1 to Part 4)’ for ‘IS 812 : 1957’.

*(Page 1, clause 4, line 3)* — Substitute ‘IS 1387’ for ‘IS 1387 : 1993’.

*(Page 1, clause 5, Table 1)* — Substitute the following for the existing:

**‘Table 1 Diameters and Tolerances**

*(Clause 5)*

| SI No. | Nominal Diameter<br>mm | Tolerance, mm |       |
|--------|------------------------|---------------|-------|
|        |                        | Plus          | Minus |
| (1)    | (2)                    | (3)           | (4)   |
| i)     | 0.6                    | 0.01          | 0.03  |
| ii)    | 0.8                    | 0.01          | 0.04  |
| iii)   | 0.9                    | 0.01          | 0.04  |
| iv)    | 1.0                    | 0.01          | 0.04  |
| v)     | 1.2                    | 0.01          | 0.04  |

**Price Group 2**

**Amendment No. 2 to IS 6419 : 1996**

**Table 1 (Concluded)**

| Sl No. | Nominal Diameter<br>mm | Tolerance, mm |       |
|--------|------------------------|---------------|-------|
|        |                        | Plus          | Minus |
| (1)    | (2)                    | (3)           | (4)   |
| vi)    | 1.4                    | 0.01          | 0.04  |
| vii)   | 1.6                    | 0.01          | 0.04  |
| viii)  | 1.8                    | 0.01          | 0.04  |
| ix)    | 2.0                    | 0.01          | 0.07  |
| x)     | 2.4                    | 0.01          | 0.07  |
| xi)    | 2.5                    | 0.01          | 0.07  |
| xii)   | 2.8                    | 0.01          | 0.07  |
| xiii)  | 3.0                    | 0.01          | 0.07  |
| xiv)   | 3.15                   | 0.01          | 0.07  |
| xv)    | 3.2                    | 0.01          | 0.07  |
| xvi)   | 4.0                    | 0.01          | 0.07  |
| xvii)  | 5.0                    | 0.01          | 0.07  |

(Page 2, clause 8.1, Table 2) — Insert the following at the end:

NOTE — Other dimensions can be as agreed between the purchaser and the manufacturer.

(Page 3, clause 10.2) — Substitute the following for the existing:

**‘10.2 Cast and Helix of Wire**

**10.2.1 Cast**

The cast of coiled filler metals shall be such as to have imparted a curvature to the filler metal so that a specimen sufficient in length to found one loop or a maximum 3 m when cut from the package and laid on a flat surface without restraint, shall form a circle or portion thereof of the diameter shown for the cast in Table 6.

**10.2.2 Helix**

The helix of coiled filler metal as executed by the ring used to determine the cast, when placed in flat surface without restraint, shall be such that the maximum distance from any point on the filler metal to flat surface shall not exceed the dimension shown for helix in Table 6.’

(Page 4, Table 5) — Delete.

(Page 5, Table 6) — Substitute the following for the existing:

**‘Table 6 Diameter of Cast and Helix**

(Clauses 10.2.1 and 10.2.2)

| Sl No. | Type of Package         | Standard Size  | Cast            | Maximum Helix |
|--------|-------------------------|----------------|-----------------|---------------|
|        |                         | mm             | mm              | mm            |
| (1)    | (2)                     | (3)            | (4)             | (5)           |
| i)     | 100 mm spool            | 1.2 and less   | 200 to 230      | 13            |
| ii)    | All except 100 mm spool | 0.8 and less   | 305, <i>Min</i> | 25            |
|        |                         | 0.9 and larger | 380, <i>Min</i> | 25            |

**Amendment No. 2 to IS 6419 : 1996**

(Page 5, clause 12) — Substitute the following for the existing:

**‘12 CHEMICAL COMPOSITION**

The chemical composition of filler rods and wires when analysed in accordance with the relevant part of IS 228 or any other established instrumental method shall be as per Table 7. In case of dispute, the procedure given in the relevant part of IS 228 shall be referee method. However, where the method is not given in IS 228 or its relevant parts, the referee method shall be as agreed to between the purchaser and the manufacturer.’

(Page 5, clause 13.1, Table) — Substitute the following for the existing:

| <i>Sl No.</i> | <i>Symbol</i> | <i>Yield Strength, Min</i> | <i>Tensile Strength, Min</i> | <i>Percentage Elongation at Gauge Length <math>5.65\sqrt{S_0}</math>, Min</i> |
|---------------|---------------|----------------------------|------------------------------|-------------------------------------------------------------------------------|
| (1)           | (2)           | MPa<br>(3)                 | MPa<br>(4)                   | (5)                                                                           |
| i)            | 50            | 400                        | 490                          | 22                                                                            |

(Page 5, clause 14.1) — Substitute the following for the existing:

**‘14.1 Chemical Composition**

The chemical composition of the filler wire or wire rod stock shall satisfy the chemical analysis as per Table 7.’

(Page 6, Table 7) — Substitute the following for the existing:

**Table 7 Chemical Composition of Filler Rods and Wires**

(Clauses 12 and 14.1)

| SI No. | IS Classification | Chemical Composition, Percent by Weight |      |                    |      |       |       |     |      |      |      |                  |      |      |      |
|--------|-------------------|-----------------------------------------|------|--------------------|------|-------|-------|-----|------|------|------|------------------|------|------|------|
|        |                   | Limits                                  | C    | Mn                 | Si   | P     | S     | Ni  | Cr   | Mo   | V    | Cu <sup>a)</sup> | Ti   | Zr   | Al   |
| (1)    | (2)               | (3)                                     | (4)  | (5)                | (6)  | (7)   | (8)   | (9) | (10) | (11) | (12) | (13)             | (14) | (15) | (16) |
| i)     | S1                | Min                                     | -    | 0.90               | 0.40 | -     | -     | -   | -    | -    | -    | -                | 0.05 | 0.02 | 0.03 |
|        |                   | Max                                     | 0.07 | 1.40               | 0.70 | 0.025 | 0.025 | b)  | b)   | b)   | b)   | 0.50             | 0.15 | 0.12 | 0.15 |
| ii)    | S2                | Min                                     | 0.06 | 0.90               | 0.45 | -     | -     | -   | -    | -    | -    | -                | -    | -    | -    |
|        |                   | Max                                     | 0.15 | 1.40               | 0.70 | 0.025 | 0.025 | b)  | b)   | b)   | b)   | 0.50             | -    | -    | -    |
| iii)   | S3                | Min                                     | 0.07 | 1.00               | 0.65 | -     | -     | -   | -    | -    | -    | -                | -    | -    | -    |
|        |                   | Max                                     | 0.15 | 1.50               | 0.85 | 0.025 | 0.025 | b)  | b)   | b)   | b)   | 0.50             | -    | -    | -    |
| iv)    | S4                | Min                                     | 0.06 | 1.40               | 0.80 | -     | -     | -   | -    | -    | -    | -                | -    | -    | -    |
|        |                   | Max                                     | 0.15 | 1.85               | 1.15 | 0.025 | 0.025 | b)  | b)   | b)   | b)   | 0.50             | -    | -    | -    |
| v)     | S5                | Min                                     | 0.07 | 1.50               | 0.50 | -     | -     | -   | -    | -    | -    | -                | -    | -    | -    |
|        |                   | Max                                     | 0.15 | 2.00 <sup>d)</sup> | 0.80 | 0.025 | 0.025 | b)  | b)   | b)   | b)   | 0.50             | -    | -    | -    |
| vi)    | S6                | No chemical requirements <sup>c)</sup>  |      |                    |      |       |       |     |      |      |      |                  |      |      |      |

<sup>a)</sup> The maximum weight percent of copper in the rod or electrode due to any coating plus the residual copper content in the steel shall be 0.50 Max,

<sup>b)</sup> These elements may be present but are not intentionally added  
Mo + Ni + Cr < 0.45,

<sup>c)</sup> For this classification, there are no chemical requirements for the elements listed, with the exception that there shall be no intentional addition of Ni, Cr, Mo or V,

<sup>d)</sup> In this classification, the maximum Mn may exceed 2.0 percent. If it does, the maximum C must be reduced by 0.01 percent for each 0.05 percent increases in Mn or part thereof.

**Amendment No. 2 to IS 6419 : 1996**

(Page 6, clause **15.1**) — Substitute the following for the existing:

‘**15.1** The lot classification shall be as per **4.1** of IS 18632 and testing schedule shall be as per **5** of IS 18632.’

(Page 7, clause **17.2**) — Substitute the following for the existing:

**‘17.2 BIS Certification Marking**

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.’

(Page 7, clause **A-1.1**) — Substitute ‘IS 2062, IS 2002, IS 3039’ for ‘IS 2062 : 1992, IS 2002 : 1992, IS 3039 : 1988’.

(Page 8, clause **A-5**) — Substitute ‘IS 1608 (Part 1)’ for ‘IS 1608 : 1995’.

(Page 9, clause **A-6**, line 3) — Substitute ‘IS 1757 (Part 1)’ for ‘IS 1757 : 1988’.

(MTD 11)

---

Publication, BIS, New Delhi